Figure 15.26 The interval subdivision method used to check intersections between two symmetric, quadratic Bézier curves (A).
(B) Compute the bounding boxes using what we know about curve's tangent at apex. Here, they overlap.
(C) We thus need to move forward, computing the tangents to the curve parallel to the axes, and use them as pivot points at which we can split the curves. Each section is now monotonic and can be interpolated with a segment: each pair of sections from the two curves will at most cross in one point.
(D) For each section in curve C1, compute the intersections using only the overlapping sections in curve 2 (then sum the results for each of the three cases).
(E) We can split each section further by halving it. Only some sub-sections will still overlap, and we can iterate steps (D-E) until they become small enough that the error in approximating a curve with a segment is within a given threshold.

15.4.5 Vertex-Vertex and Edge-Vertex Intersections

So far, we have delayed the discussion about validating an embedding by checking that no pair of vertices share the same position, and that no edge crosses a vertex.

There are many ways to draw edges, as we have seen, but there are also many ways to draw vertices: they can be punctiform, they can be drawn as circles, but they can also